For loop :

syntax :

```
for (initialization ; condition ; update){
    // statements
}
```

(1) example :

```
public class ForLoopExample {
    public static void main (String[] args ){
        for (int i = 1; i <= 5; i++){
            System.out.println (" Count : "+ i);
        }
    }
}
```

* initialization is done as per the preferences or convinience.
* println is for specific or next line.
* print is for continuous line
* Update can be done as per need (decrement (--) / increment (++))
* If condition fails then the loop tends to stop.

(2) Example :

```
public class Example {
    public static void main ( String [] args) {
        int sum = 0;
        for ( int i = 1 ; i<=10 , i++) {
            sum += i;
        }
        System.out.println ("Sum=" + sum);
    }
}
```

* if written inside it gives step-by-step output.

* Written outside the loop for the final output

__/__/__

* Odd / Even with condition

→ with if :
  if (i % 2 != 0) → prints odd

→ without if :
  for (int i = 1; i <= 10 ; i += 2) → prints odd
  for (int i = 2; i <= 10 ; i += 2) → prints even

* Pattern printing :

Example :

```
public class Pattern {
    public static void main (String[] args){
        for (int i = 1; i<=5 ; i++) {              ← outer loop
            for (int j = 1; j <= i ; j++) {        ← inner loop
                System.out.print ("* ");
            }
            System.out.println();
        }
    }
}
```

output:

```
*
* *
* * *
* * * *
* * * * *
```

if i = 1 j = 1;
i gets incremented to 2
i is for ~~columns~~ row
j is for column

★ 2 to ~~10~~ 5 multiplication tables :

```
public class MultiplicationTable {
    public static void main (String [] args) {
        for (int i = 2, i<=5 ; i++) {
            for (int j = 1 ; j<=10 ; j++) {
                System.out.print ( i+ "x" +j+ "="
                    + i*j+ "\n");
            }
            System.out.println ( );
        }
    }
}
```

nested for loops

→ "\t" is for tab / indentation

★ 5 x 5 matrix pattern :

O/P :

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	25

```
public class MultiplicationTable {
    public static void main( String[] args) {
        int sum = 0 ;
        for (int i = 1 ; i<=5 ; i++) {
            for (int j = 1 ; j<=5 ; j++) {
                sum ++;
                System.out.print( sum + "\t") ;
            }
            System.out.println ( ) ;
        }
    }
}
```

nested loops for matrix formation

~~to~~ sum or increment by 1